ഗണിത ശാസ്ത്ര വിഭാഗം
DEPARTMENT OF MATHEMATICS
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
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DM/132/2026-27/ **29.06.2026**

This is to certify that **Ms. Malavika M, Srilekshmi, Inchipulluvila, Pappanamcode P.O, Thiruvananthapuram, PIN-695018** successfully completed the **120-hour Summer Internship Programme (June 2026)** titled "Advanced mathematical Foundations of Machine Learning and Deep Learning Optimization" under the guidance of **Dr. Linu Pinto**, Assistant Professor, Department of Mathematics, CUSAT from 1 June 2026 – 29 June 2026 (120 hours) with additional training by expert resource persons through lectures, tutorials, group discussions, and assessment tests. Internship work was the development of LaTeX documentation on the following topics and respective scratch Python implementation of algorithms.

The participant underwent training in:

- **Linear Algebra & Geometry:** matrix operations, four fundamental subspaces, LU & QR decomposition, eigenvalues, SVD, PCA, and optimal low-rank approximation.
- **Optimization algorithms:** Gradient Descent, Momentum, AdaGrad, RMSProp, Adam, Newton's Method, and Quasi-Newton methods (BFGS, L-BFGS).
- **Backpropagation (BP) algorithm** on deep learning architectures, with mathematical derivation and from-scratch Python code implementation.
- **Understanding on the Building and training** on MLP, RNN, LSTM, CNN, and foundational Transformer architectures without advanced Deep learning libraries like TensorFlow Keras and PyTorch.
- **Understanding** how deep learning algorithms are developed thereby preparing participants to adapt independently to the evolving AI landscape.

We appreciate her consistent and active involvement in the programme.

29/06/26

Dr. LINU PINTO
Assistant Professor
Department of Mathematics
Cochin University of Science and Technology
Cochin - 682022, Kerala, India

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF MATHEMATICS
KOCHI-22

29/6/26

PROFESSOR AND HEAD
Dr. SHERY FERNANDEZ
PROFESSOR & HEAD
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DEPARTMENT OF MATHEMATICS
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY (CUSAT)
Kochi – 682 022, Kerala, India

INTERNSHIP PERFORMANCE GRADE CARD

Summer Internship Programme – Advanced Mathematical Foundations of ML & Deep Learning – June 2026

Student Name: Malavika M.

A. CRITERION-WISE EVALUATION

Sl.	Evaluation Criterion	Grade
1	Depth of understanding of core concepts (Linear Algebra, Optimization, Deep Learning)	O
2	Quality and completeness of LaTeX-typeset notes (mathematical notation, equations, structure)	O
3	Successful implementation of architectures (MLP, RNN, LSTM, CNN, Transformer)	O
4	Ability to explain geometric intuition behind matrix operations and transformations	O
5	Understanding of backpropagation flow through deep architectures	O
6	Evidence of self-study and exploration	O
7	Logical flow and coherence of the internship report (structure, argumentation, section organization)	O
8	Ability to communicate complex mathematical ideas with clarity and appropriate examples	O
9	Scores in assessment tests conducted during the programme	O
10	Participation in group discussions and interactions	O
	TOTAL	

B. GRADE SCALE

Grade	Descriptor
Outstanding (O)	The student demonstrates exceptional mastery of the concepts, possessing the depth and maturity to independently pursue and contribute to research in Machine Learning and Deep Learning
Excellent (A)	The student possesses a strong conceptual foundation, demonstrating the ability to learn independently and advance further in AI, data science, and related professional domains.
Good (B)	The student has a sufficient conceptual foundation to pursue further study in this field through internships, academic courses, or research programmes.
Satisfactory (C)	Foundational understanding of the topics covered; further coursework and practice are recommended to strengthen depth.
Unsatisfactory (U)	Needs substantial further training in the foundational concepts before progressing in this field.

C. OVERALL GRADE CALCULATION

Overall Grade: O

Remarks: Outstanding.

Signature of Evaluator:

Date: 29/06/26

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